

NOTES AND COMMENTS

A FAIRNESS JUSTIFICATION OF UTILITARIANISM

PAOLO GIOVANNI PIACQUADIO

Department of Economics, University of Oslo and Statistics Norway

Differences in preferences are important to explain variation in individuals' behavior. There is, however, no consensus on how to take these differences into account when evaluating policies. While prominent in the economic literature, the standard utilitarian criterion is controversial. According to some, interpersonal comparability of utilities involves value judgments with little objective basis. Others argue that social justice is primarily about the distribution of commodities assigned to individuals, rather than their subjective satisfaction or happiness. In this paper, we propose and axiomatically characterize a criterion, named *opportunity-equivalent utilitarian*, that addresses these claims. First, our criterion ranks social alternatives on the basis of individuals' ordinal preferences. Second, it compares individuals based on the fairness of their assignments. *Opportunity-equivalent utilitarianism* requires society to maximize the sum of specific indices of well-being that are cardinal, interpersonally comparable, and represent each individual's preferences.

KEYWORDS: Utilitarianism, ordinal preferences, fairness, opportunities.

1. INTRODUCTION

UTILITARIANISM IS ARGUABLY the most prominent theory of distributive justice. In the context of income distributions, utilitarian welfare is generally measured by

$$W = \sum_{i \in N} U_i(y_i), \quad (1)$$

where U_i is individual i 's utility function, that is, a numerical representation of preferences defined over i 's income y_i which is interpersonally comparable and unique up to a positive affine transformation. It is standard to assume that the functions $(U_i)_{i \in N}$ are equal across individuals as individuals are identical; increasing as individuals prefer more income to less; and concave by the law of “decreasing marginal utility” (the effect on U_i of an additional unit of income decreases with the level of income).

The intuitive appeal of (1) may be also explained by an alternative “non-utility-based” justification, which relies on fairness arguments. Society *chooses* the functions $(U_i)_{i \in N}$ to aggregate the conflicting interests of individuals and measure social welfare. These functions are set equal across individuals to ensure that each individual is treated anonymously (permuting their incomes leaves social welfare unchanged); increasing to yield efficiency (if individuals are better off, social welfare is higher); and concave by Dalton (1920)'s “principle of transfer” (a transfer from a poorer individual to a richer one increases inequality and reduces social welfare).

Paolo Giovanni Piacquadio: p.g.piacquadio@econ.uio.no

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Crucially, these two justifications support the same policy recommendations when the concavity due to decreasing marginal utilities is the same as the concavity introduced for inequality aversion. Thus, when evaluating income distributions, some versions of “utilitarianism” need not be divorced from the ordinalist foundations of microeconomics. In this paper, we establish a similar result for the evaluation of distributions of commodities to individuals with (possibly) different preferences. Note that our use of the term “utilitarianism” is broader than usual. Standard utilitarian welfare is the sum of each individual’s utility. Our “utilitarian” criteria are the sum of *some* indices of well-being that—as utility functions—are cardinal, interpersonally comparable, and represent each individual’s preferences. While these indices may coincide with individuals’ utility functions, we propose these be selected based on fairness arguments.

Our fairness justification for the multicommodity analog of (1) is consistent with two important critiques raised against standard utilitarianism. Following Robbins (1932) and Samuelson (1947), several economists charge utilitarianism with a major challenge: interpersonal comparisons of utilities rely on value judgements with little or no justification; they thus suggest individuals’ information to be restricted to ordinal preferences, as in our analysis. More recently, economists and philosophers have criticized the view that interpersonal comparisons ought to be made in terms of subjective satisfaction or happiness. Rawls (1971) and Sen (1992) suggested comparing individuals by an index of the resources or capabilities individuals are assigned. Dworkin (1981) and Fleurbaey and Maniquet (2011) observed that this approach is inconsistent with the Pareto principle and, thus, disregards valuable information about what is good for each individual.¹ Following Dworkin and Fleurbaey and Maniquet, we propose that interpersonal comparisons be made in terms of individuals’ assignments, while taking into account that individuals may value assignments differently.

The social concern for fairness is here captured by three intuitive principles of justice.

Possibility of trade-offs tells that, in specific situations, society can compensate a loss for one individual with a gain for some other individual. To illustrate, let each individual but individual i be assigned the zero bundle (by assumption the worst assignment). Then, for each individual $j \neq i$, there is a socially indifferent alternative that assigns a nonzero bundle only to j . In other words, the loss that individual i experiences—when her bundle is reduced to zero—can be compensated by a gain of any other individual—assigning her some positive bundle instead of zero. This requirement limits the influence that each individual can exert on the social ranking.

Nondiscrimination means that no individual is considered more deserving than any other.² We first illustrate the axiom in the context of income distributions. Consider a distribution which assigns income $y_i > 0$ to individual i and no income to all other individuals. Another income distribution assigns a lower income to individual j , that is,

¹Differences in individuals’ preferences matter for social welfare. These need to be accounted for to explain individual and aggregate economic behavior (Dubois, Griffith, and Nevo (2014), Heathcote, Storesletten, and Violante (2014)) and to correctly evaluate welfare effects of policies (Kirman (1992)). In some contexts, however, the preferences of individuals might not be considered legitimate for society (as in the presence of *schadenfreude*, overconfidence, status quo bias, inattention, myopia, addiction, etc.). In other contexts, information about preferences might be subject to strategic manipulation (in particular in small-group frameworks). To obviate these problems, society might replace individuals’ true or revealed preferences by preferences that society considers legitimate. Our results are unaffected by such a change.

²By assumption, individuals may differ only with respect to preferences and assignments. We thus exclude from our analysis further differences between individuals—that is, household size, special needs, etc.—which would justify considering some individuals more deserving than others. The nontrivial extension to such cases requires identifying some kind of equivalent scales and is left to future research.

$y_j < y_i$, and no income to all other individuals (including i). If society prefers the latter distribution, individual i is discriminated against: society would rather assign less to j than more to i . The axiom rules out this type of social preferences. In a multicommodity framework, society needs to account for differences in preferences. Consider all allocations at which individual i is indifferent between her assignment x_i and some bundle $\bar{x}_i$, while all other individuals are assigned the zero bundle. Then, i is discriminated against if for each such allocation, society prefers to assign a smaller bundle to some individual $j \neq i$, that is, $x_j < \bar{x}_i$, and the zero bundle to all others (including i).³ Such discrimination is not permitted.

Equal-preference transfer introduces a social concern for inequality. This axiom is inspired by Dalton's transfer principle. Consider two individuals, i and j , with the same preferences.⁴ At allocation x , individual i is assigned a bundle on a higher indifference curve than individual j . At allocation x' , the difference between their assignments is reduced by a transfer of commodities from i to j , while all other individuals are unaffected. Then, society considers x' at least as desirable as x .⁵

Three further requirements are standard. *Efficiency* requires social welfare to increase when some individuals are better off and no one is worse off. *Continuity* says that small changes of the allocation should not cause large jumps in the level of social welfare. *Separability* requires the ranking of two allocations to be independent of the assignment of an individual who is assigned the same bundle in both allocations.

The above axioms uniquely characterize the class of "opportunity-equivalent utilitarian" welfare criteria. Each member of this class is identified by two objects.

The first object is a family of nested opportunity sets, where an opportunity set is a set of consumption bundles. The family can be parametrically described. The larger the value of the parameter, the larger the opportunity set to which it is associated. This value identifies the well-being level of an individual who is: (i) either assigned (one of) her most preferred bundle in the opportunity set associated with this value; or (ii) any other bundle she finds equally desirable.⁶ Whenever two individuals are assigned their preferred bundles from the same opportunity set, they are regarded as achieving the same well-being level. If one is assigned her preferred bundle from a larger opportunity set, she is regarded as achieving a higher well-being level. The functions that, for each individual, associate the above-defined well-being level to each assignment constitute the "opportunity-equivalent" representations of their preferences. These representations are interpersonally comparable, but only ordinal: the values associated to each opportunity set are unique up to any order-preserving transformation. Cardinal meaning is achieved when these functions are subjected to a common "cardinalizing function," endogenously singled out by the axioms. It is the least concave function which ensures concavity of each individual's numerical representation of preferences.⁷

³Here $x_j < x_i$ means that, in each dimension, x_j is not larger than x_i and that $x_j \neq x_i$.

⁴Individuals' preferences are assumed to be convex, ensuring that our transfer axiom does not conflict with efficiency.

⁵In a multicommodity setting, it seems natural for society to value progressive transfers between *any* two individuals, when one is assigned more of each commodity. When individuals have different preferences, however, this axiom clashes with the Pareto principle (Fleurbaey and Trannoy (2003)).

⁶Here, we measure each individual's well-being by a comparable index of her welfare that respects her preferences: "when preferences are self-interested, well-informed, and undistorted, ... it is sensible for those seeking to promote welfare to employ methods of appraising policies... that rely on information concerning preference satisfaction" (Hausman (2012, p. 102)).

⁷To illustrate, consider preferences over income. Any increasing function (convex, concave, or neither convex nor concave) represents monotonic preferences over income. Moreover, any increasing transformation of

The second object is an increasing and concave function that is applied to the profile of well-beings. This function reflects inequality aversion of the society and is standard for “generalized” versions of the utilitarian criterion (Fleming (1952)).

Beyond the degree of inequality aversion, *opportunity-equivalent utilitarian* criteria accommodate a large variety of views about justice. The shape of opportunity sets reflects the ethical importance and substitutability of commodities. Among the special cases: money metric representations of preferences (McKenzie (1957) and Samuelson and Swamy (1974)) emerge when each opportunity set consists of the bundles that can be purchased (for fixed prices) at some wealth level;⁸ ray representations of preference (Pazner (1979)) emerge when each opportunity set consists of the bundles that are not larger than some fraction or multiple of a fixed reference bundle. Moreover, commodities may represent resources (Dworkin (1981), Fleurbaey and Maniquet (2011)), or functioning and capabilities (as suggested by Sen (1992)), or a combination of primary goods and resources (Rawls (1971)).

Our results also shed light on the dispute on ordinalist welfare analysis. Bergson (1938), and later Samuelson (1947), suggested that welfare criteria be based only on individuals’ indifference maps.⁹ In the late 1970s, the hope to construct Bergson–Samuelson social welfare functions succumbed to the robustness of Arrow (1963)’s theorem (Kemp and Ng (1976) and Parks (1976)). The economic community considered settled the controversy and concluded that utility information is necessary to pass from individual tastes to social preferences, if social preferences are to reflect individual preferences without being dictatorial.

Few economists rejected this conclusion. Samuelson (1977) criticized the ethical appeal of the set of axioms imposed by Kemp and Ng (1976), but only provided an example proving that “reasonable” ordinalist criteria exist. Motivated by similar critiques, Mayston (1974, 1982) suggested a set of rules to generate a large class of ordinalist welfare criteria. Unfortunately, the extreme richness of his class provides little help to welfare analysis. Pazner (1979), and in particular Fleurbaey and Maniquet (2011), provided much sharper welfare criteria. Within specific economic domains, they showed how to construct ordinalist criteria based on fairness principles similar to ours. However, by disregarding some information about individuals’ preferences, their axioms force society to be infinitely averse to inequality: their *maximin* type of welfare criteria says that no sacrifice for the worst-off

this function represents the same preferences, meaning that such representation is only ordinal. In contrast, a least concave representation of preferences over income is a positive affine function of income. Moreover, least concave representations are unique up to positive affine transformations and, thus, are cardinal (see Debreu (1976)). In our analysis, two differences emerge: (i) preferences are defined over multicommodity spaces; and (ii) the cardinalizing function ensures that the representations of preferences are “least jointly concave,” that is, minimally concave for all individuals at the same time. These differences, however, do not affect cardinality: the least jointly concave representations of preferences are unique up to positive affine transformations.

⁸Interestingly, our aggregation of money metrics is immune to the critique by Blackorby and Donaldson (1988). They showed that a society maximizing the sum of money metrics may value positively a transfer from a worse-off individual to a better-off individual (even if these had the same preferences). In our framework, money metrics may guide society in making interpersonal comparisons of well-being. However, the cardinal indices used in our welfare criterion obtain as a transformation of the money metric representations. This transformation ensures that each individual’s numerical representations of preferences is concave and, consequently, that society does not endorse regressive transfers. On the measurement of social welfare by the sum of money metrics, see also the characterization by Bosmans, Decanq, and Ooghe (2016).

⁹As Arrow (1963, p. 109) wrote, the ordinal informational basis is a grounding feature and key contribution of Bergson’s approach: “It is the great merit of Bergson’s 1938 paper to have carried the same [ordinalist] principle into the analysis of social welfare. The social welfare function [or welfare criterion] was to depend only on indifference maps...”

is acceptable, independently of how large the benefits for all other individuals in society are. *Opportunity-equivalent utilitarianism* avoids the above limitations. On the one side, it provides enough guidance to deliver clear-cut policy recommendations. On the other side, it is flexible enough to avoid society giving full priority to the worst off.

The crucial difference with Arrow (1963), Pazner (1979), and Fleurbaey and Maniquet (2011) lies in the information on individuals' preferences that is considered relevant for society. Arrow's "independence of irrelevant alternatives" requires the ranking of two alternatives to be unaffected by all changes of preferences, provided the ranking of the assigned bundles remains unchanged. Thus, social preferences can only depend on how individuals rank each pair of bundles. Pazner and Fleurbaey and Maniquet adopted a weaker requirement: "independence of non-indifferent alternatives" requires the ranking of two alternatives to be unaffected by all changes of preferences, provided the indifference curves through the assigned bundles remain unchanged. Thus, social preferences can depend on individuals' indifference curves through each pair of bundles, allowing for the construction of interpersonally comparable indices of well-being. The above-mentioned money metric and ray representations of preferences are examples of indices of well-being that require information about individuals' indifference curves and, thus, violate "independence of irrelevant alternatives." In our analysis, we drop Arrow's requirement completely and do not disregard any information about individuals' preferences. Our results clarify that the additional information allows identifying indices of well-being with cardinal meaning, thus avoiding the maximin type of criteria obtained in previous literature.¹⁰

The remainder of the paper is organized as follows. Section 2 presents the model and the axioms. Section 3 contains the main results. Section 4 concludes. A domain maximality result and the independence of the axioms are presented in the online Supplemental Material (Piacquadio (2017)).

2. THE FRAMEWORK AND THE AXIOMS

2.1. The Framework

A society consists of a finite set of individuals $N \equiv \{1, \dots, n\}$ with $n \geq 3$. Each individual $i \in N$ is characterized by a preference relation R_i ; R_i is a weak order on the ℓ -dimensional Euclidean commodity space $X \equiv \mathbb{R}_+^\ell$, with ℓ finite.¹¹ The strict preference and indifference relations induced by R_i are denoted by P_i and I_i . Each preference relation R_i is strongly monotonic, continuous, and admits a concave representation. For each individual $i \in N$ with preferences R_i , there is an individual $j \in N \setminus \{i\}$ with the same preferences $R_j = R_i$. We discuss these assumptions below.

An allocation $x \equiv (x_1, \dots, x_n) \in X^n$ assigns a bundle x_i to each individual $i \in N$. For each $i \in N$ and each $x_i \in X$, let $e^i(x_i) \in X^n$ be the allocation that assigns x_i to individual i and the zero bundle to all other individuals. A social ranking, denoted $\succsim$, is a weak order of allocations. For each pair of allocations $x, x' \in X^n$, $x \succsim x'$ means that x is at least as

¹⁰Recently, Fleurbaey and Tadenuma (2014) proposed a different way to escape maximin type of criteria. They combined "independence of non-indifferent alternatives" with a weakening of the equity principle, that is, a multicommodity progressive transfer among individuals with the same and homothetic preferences. The drawback is that society might want to redistribute commodities from a worse-off individual to a better-off individual, when these have the same, but non-homothetic, preferences. As we discuss later, homothetic preferences play a special role and lead to a simplified representation of our welfare criterion.

¹¹ $\mathbb{R}_+^\ell$ denotes the nonnegative orthant of $\mathbb{R}^\ell$. Vector inequalities are denoted $\geq$, $>$, and $\gg$.

socially desirable as x' . The strict social preference and indifference relations induced by $\succsim$ are denoted by $\succ$ and $\sim$.

REMARK 1: We adopt the single-profile approach, meaning that the social ranking is defined only for the specific society N in which each individual $i \in N$ has preferences R_i . The multi-profile approach addresses the mapping that associates a social ranking to each possible society. Clearly, the entire mapping obtains by repeating the single-profile approach for each possible society. The only difference, then, is that interprofile conditions—such as Arrow’s “independence of irrelevant alternatives”—do not have place in the single-profile approach (see d’Aspremont and Gevers (2002)).

REMARK 2: The commodity space X can be generalized to any convex subset of $\mathbb{R}_+^\ell$ such that there exists a bundle that is considered worst by all individuals. When $X = \mathbb{R}_+^\ell$, this worst bundle is the zero vector. The results do not extend to discrete commodity spaces or to the analysis of public goods.

REMARK 3: A preference relation that admits a concave numerical representation is also convex, but the reverse is not true (even if preferences are continuous). Our axioms are compatible only if preference relations are representable by concave functions (see Piacquadio (2017)). Nevertheless, this is not a very restrictive assumption as smooth and strictly convex preference relations defined on compact sets admit concave numerical representations (Mas-Colell (1985)). We leave to future research the extension of our results to preferences that are not complete, transitive, continuous, or convex (for a recent contribution in this direction, see Argenziano and Gilboa (2017)).

REMARK 4: We require each preference relation in society to be shared by at least two individuals. This richness assumption ensures that a later-introduced fairness axiom—involving exclusively individuals with the same preference relation—is non-vacuous. While simplifying the exposition, the same conclusions can be derived without this assumption. In the multi-profile approach, it can be replaced by a replication invariance axiom. The social ranking should be unchanged when comparing two allocations for a specific society and when comparing the replicas of these allocations for the replicas of these societies.

2.2. Axioms

The next three axioms are standard in the literature. First, if each individual finds her assignment at $x \in X^n$ at least as desirable as her assignment at $x' \in X^n$ and some individuals prefer their assignment at x to their assignment at x' , then x is socially preferred to x' .

Efficiency: For each pair $x, x' \in X^n$, $x_i R_i x'_i$ for each $i \in N$ and $x_i P_i x'_i$ for some $i \in N$ implies $x \succ x'$.

Next, small changes in allocations do not cause large jumps in social welfare.

Continuity: For each $x \in X^n$, the sets $\{x' \in X^n | x' \succsim x\}$ and $\{x' \in X^n | x \succsim x'\}$ are closed.

Moreover, the ranking of two alternatives is independent of the bundle assigned to an individual who is unconcerned by the choice.¹²

¹²The following notation is standard: for each $x \in X^n$, each $i \in N$, and each $a_i \in X$, let $(a_i, x_{-i}) \in X^n$ be the allocation that assigns a_i to i and x_j to each $j \in N \setminus \{i\}$.

Separability: For each pair $x, x' \in X^n$, if there is $i \in N$ such that $x_i = x'_i \equiv a_i$, then for each $b_i \in X$,

$$(a_i, x_{-i}) \succsim (a_i, x'_{-i}) \iff (b_i, x_{-i}) \succsim (b_i, x'_{-i}).$$

Next, the social concern for fairness is embodied in the following three axioms.

The first limits the influence that any individual can exert on the social ranking. Let $i \in N$ and $x_i \in X \setminus \{0\}$. We require that, for each $j \in N$, there exists a bundle $x_j \in X \setminus \{0\}$ such that $e^j(x_j)$ is socially indifferent to $e^i(x_i)$. Society can then trade off the well-being of any two individuals, ruling out dictatorial welfare criteria.¹³

Possibility of Trade-offs: For each pair $i, j \in N$ and each $x_i \in X \setminus \{0\}$, there exists $x_j \in X \setminus \{0\}$ such that $e^j(x_j) \sim e^i(x_i)$.

The next axiom prevents a certain form of discrimination among individuals: it precludes society to consider some individual more deserving than the others.¹⁴ Let $i \in N$ and $x_i^* \in X$. Then, i is discriminated against if, for each $x_i \in X$ such that $x_i I_i x_i^*$, society is worse off at allocation $e^i(x_i)$ than at any allocation $e^j(x_j) \in X^n$ which assigns a smaller bundle $x_j < x_i$ to some individual $j \in N \setminus \{i\}$. In other words, individual i is discriminated against if there is no bundle on her indifference curve through x_i^* which society would prefer i to consume instead of giving less to some other individual. Such discrimination is prevented here for each $i \in N$ and each $x_i^* \in X$.

Nondiscrimination: For each $i \in N$ and each $x_i^* \in X$, there exists $x_i \in X$ with $x_i I_i x_i^*$ such that, for each $j \in N \setminus \{i\}$ and each $x_j \in X$, $e^j(x_j) > e^i(x_i)$ implies $x_j \not\prec x_i$.

With one commodity only, say income, *nondiscrimination* is weaker than anonymity. Society is indifferent to income permutations on the subset of distributions which assign a positive income to at most one individual.

The last axiom is related to the Pigou–Dalton transfer principle. Interpreting a justice concern expressed by Pigou (1912), Dalton (1920) suggested that a progressive transfer from a richer to a poorer individual (provided the richer/poorer relation is not reversed) leads to a more desirable distribution of income. Similarly to Fleurbaey and Maniquet (2011), we extend the transfer to multiple commodities and restrict it to individuals with the same preferences.

Equal-Preference Transfer: For each pair $j, k \in N$ such that $R_j = R_k \equiv R_0$, if there exist a pair $x, x' \in X^n$ and $\alpha \geq 0$ such that:

- (i) $x_j - \alpha(x_j - x_k) = x'_j$, $R_0 x'_k = x_k + \alpha(x_j - x_k)$;
 - (ii) for each $i \in N \setminus \{j, k\}$, $x_i = x'_i$;
- then $x' \succsim x$.

¹³This axiom is inspired by “weak anonymity” introduced in d’Aspremont (1985) (see also d’Aspremont and Gevers (2002)). Weak anonymity requires that for each pair of individuals $i, j \in N$ there exist two allocations $x^i, x^j \in X^n$ such that: $x^i P_i x^j$; $x^j P_j x^i$; and $x^i I_k x^j$ for each $k \neq i, j$. Our axiom is stronger: we ask society to be able to trade off any bundle assigned to i only for some bundle assigned to j only.

¹⁴This is related to the problem of expensive tastes, summarized by Sen (1979, p. 203) as follows: “if person A as a cripple gets half the utility that the pleasure-wizard person B does from any given level of income, then in the pure distribution problem between A and B the utilitarian would end up giving the pleasure-wizard B more income than the cripple A.” In our approach, information is restricted to individuals’ ordinal preferences: no information is available about the capacity of individuals to transform resources into utilities. Thus, our axiom impedes society to arbitrarily consider some individuals “cripples” and others “pleasure-wizards.” Of course, there are settings—ruled out here—where this discrimination is morally justified: for instance, when households consist of different number of individuals or in the presence of different needs or disabilities.

The axiom reads as follows. Individuals $j, k \in N$ have the same preferences $R_j = R_k \equiv R_0$. At allocation $x \in X^n$, individual j is assigned a more desirable bundle than individual k according to R_0 . At $x' \in X^n$, the difference between assignments is reduced by a transfer that is proportional to $x_j - x_k$; importantly, j remains at least as well off as k after the transfer, that is, $x'_j R_0 x'_k$. Since all other individuals are unaffected, the after-transfer allocation has less inequality and is considered at least as socially desirable as the initial one.

3. THE MAIN RESULT

In this section, we define our welfare criteria and present the characterization result.

3.1. The Opportunity-Equivalent Utilitarian Criterion

The criterion to be characterized can be numerically represented by the sum of specific indices of well-being. It follows that these indices need to be comparable with respect to gains and losses of different individuals and unique up to positive affine transformations. We construct these indices in two steps. The first step identifies how to make interpersonal comparisons: society considers individuals equally well-off when they are assigned “equivalent opportunities.” The second step defines cardinal indices of well-beings and builds on the idea of least-concave representations of preferences, introduced by Debreu (1976).

An *opportunity mapping* C identifies a family of nested sets of bundles (Thomson (1994)). For each $\lambda \in \mathbb{R}_+$, $C(\lambda) \subseteq X$ is the *opportunity set* of “size” λ : it is nonempty, compact, and satisfies free disposal (for each pair $c, c' \in X$ with $c \geq c'$, $c \in C(\lambda) \Rightarrow c' \in C(\lambda)$). Also, $C(0) = \{0\}$ and $\lim_{\lambda \rightarrow \infty} C(\lambda) = X$. Moreover, C is continuous and “strongly monotonic” (for each pair $\lambda, \lambda' \in \mathbb{R}_+$ with $\lambda' > \lambda$ and each $x \in C(\lambda)$, there exists $x' \gg x$ such that $x' \in C(\lambda')$). Let $\mathcal{C}$ be the set of opportunity mappings. For each $i \in N$ and each $\tilde{X} \subseteq X$, let $m_i(\tilde{X})$ denote the set of i ’s most preferred bundles in $\tilde{X}$.

Consider an opportunity mapping $C \in \mathcal{C}$. Individual i is assigned a bundle $x_i \in X$. Select $\lambda_i \in \mathbb{R}_+$ such that individual i is indifferent between her preferred alternative in the set $C(\lambda_i)$ and x_i or, more formally, satisfies $x_i I_i z_i$ with $z_i \in m_i(C(\lambda_i))$. Then, λ_i can be interpreted as a measure of the well-being of individual i at x_i . The role of the opportunity mapping C for interpersonal comparisons of well-being is illustrated in Figure 1. Individual i is assigned the bundle x_i , which is for her indifferent to her preferred alternative in $C(\lambda_i)$. Individual j is assigned the bundle x_j , which is for her indifferent to her preferred alternative in $C(\lambda_j)$. When comparing the assignments x_i and x_j , what matters is the size of these opportunity sets $C(\lambda_i)$ and $C(\lambda_j)$. As $\lambda_i > \lambda_j$, individual i is considered better off than individual j . Importantly, such comparisons are not affected by order-preserving transformations of the parameter λ identifying the size of the opportunities.¹⁵

Formally, for each $i \in N$, the *opportunity-equivalent well-being function relative to* $C \in \mathcal{C}$ is the function $U_i^C : X \rightarrow \mathbb{R}$ defined by setting, for each $x_i \in X$,

$$U_i^C(x_i) \equiv \lambda_i \iff x_i I_i z_i \text{ with } z_i \in m_i(C(\lambda_i)).$$

¹⁵Conversely, the choice of $C \in \mathcal{C}$ matters for the identification of the worse-off individual. Nevertheless, it is not identified by the characterization result and is a free ethical choice. This is unsurprising: the shape of opportunity sets defines the ethical importance and substitutability of commodities. Even restricting to opportunities for welfare delimited by a linear function (i.e., budget sets), this would require specifying the relative prices of commodities: this cannot be chosen in an abstract setting, where commodities could represent consumption goods, basic rights, or capabilities.

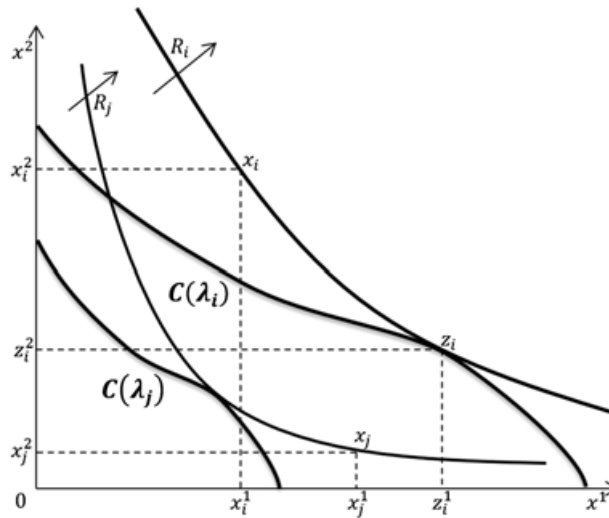


FIGURE 1.—The opportunity-equivalent well-being function.

As we shall see, society makes interpersonal comparisons through the functions $(U_i^C)_{i \in N}$ for some choice of opportunity mapping $C \in \mathcal{C}$. For each $i \in N$, U_i^C is a continuous numerical representation of preferences R_i . Given an opportunity mapping $C \in \mathcal{C}$ and the corresponding opportunity-equivalent well-being functions $(U_i^C)_{i \in N}$, we can now construct our cardinal indices of well-being. Define Ψ^C as the set of all real-valued functions $\psi : \mathbb{R} \rightarrow \mathbb{R}$ such that, for each $i \in N$, $\psi \circ U_i^C$ is a concave representation of preferences R_i . Let $\Phi^C \subset \Psi^C$ be the set of least joint concave transformations: $\phi \in \Phi^C$ implies that for each $\psi \in \Psi^C$, ψ is more concave than ϕ .¹⁶ Importantly, if $\phi \in \Phi^C$, then any positive affine transformation of ϕ also belongs to Φ^C . Then, for any $\phi \in \Phi^C$, society considers the transformed opportunity-equivalent well-being functions $(\phi \circ U_i^C)_{i \in N}$ as cardinal and interpersonally comparable indices of well-being.¹⁷ We next show that such transformations exist. The proof is provided in the [Appendix](#).

LEMMA 1: *For each opportunity mapping $C \in \mathcal{C}$, the set Φ^C is nonempty.*

The social welfare criterion can now be defined. A social ranking $\succsim$ is *opportunity-equivalent utilitarian* if there exist $C \in \mathcal{C}$, $\phi \in \Phi^C$, and an increasing and concave function f , such that $\succsim$ is represented by the function $W : X^n \rightarrow \mathbb{R}$, defined by setting, for each

¹⁶A function $s : \mathbb{R} \rightarrow \mathbb{R}$ is more concave than a function $s' : \mathbb{R} \rightarrow \mathbb{R}$ if there exists a real-valued and concave transformation $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $s = g \circ s'$. Least concave numerical representations of preferences were studied in [Debreu \(1976\)](#) and [Kannai \(1977\)](#). These were later applied to measure risk ([Kihlstrom and Mirman \(1981\)](#)), inequality ([Atkinson and Bourguignon \(1982\)](#), [Bosmans, Decancq, and Ooghe \(2015\)](#)), and poverty ([Atkinson \(2003\)](#)).

¹⁷To preserve interpersonal comparability of the opportunity-equivalent well-being functions, we derive the least joint concave representations of preferences: the transformation ensuring least concavity of the representations is jointly imposed on all individuals. A related idea was discussed in [Mandler \(2006\)](#): to compromise between ordinality and cardinality of utility information, he suggested taking as primitives all the increasing and concave transformations of ordinally comparable utilities and developed a partial order based on joint concavity.

$x \in X^n$,

$$W(x) = \sum_{i \in N} f \circ \phi \circ U_i^C(x_i).$$

Each addend requires the specification of three objects:

(i) U_i^C is the representation of preferences that society assigns to each individual $i \in N$ to (ordinally) compare them. These functions are uniquely determined by the opportunity mapping $C \in \mathcal{C}$. At bundle x_i , the level of well-being achieved by i is the size λ_i of the opportunity set $C(\lambda_i)$ that makes her indifferent between her assignment x_i and her preferred bundle in $C(\lambda_i)$.

(ii) $\phi \in \Phi^C$ ensures that the representation of preferences $\phi \circ U_i^C(x_i)$ has cardinal meaning. As Φ^C consists of all positive affine transformations of ϕ , the choice of $\phi \in \Phi^C$ is irrelevant for the social ranking.¹⁸ The function ϕ guarantees that for each individual $i \in N$, $\phi \circ U_i^C$ is concave. Furthermore, it is among the least concave transformations that do so.

(iii) f is a real-valued and concave function. The concavity of f reflects social inequality aversion. The more concave f is, the more society wants to redistribute from individuals who are considered better off to individuals who are considered worse off. At the limit, as concavity tends to infinity, society assigns absolute priority to the individual who is considered worst off.

In the one-commodity case—say income—the opportunity-equivalent well-being functions are identical across individuals. An opportunity-equivalent well-being function U^C is any increasing (concave or convex or neither concave nor convex) function of income. A least concave representation would be the income itself, or any positive affine transformation of income. Thus, the least joint concave transformation ϕ is (an affine transformation of) the inverse of U^C : it ensures that the least jointly concave well-being functions are positive affine functions of income and, thus, have cardinal meaning. The family of criteria we characterize is then the sum of any concave transformation—identified by the function f —of each individual's income level. This solution corresponds to the “non-utility-based” justification of (1), discussed in the [Introduction](#).

3.2. The Characterization Result

Our main result is that a social ranking satisfies the axioms introduced above if and only if it is *opportunity-equivalent utilitarian*. The proof is presented in the next subsections.

THEOREM 1: *A social ranking satisfies efficiency, continuity, separability, possibility of trade-offs, nondiscrimination, and equal-preference transfer if and only if it is opportunity-equivalent utilitarian.*

This result shows that the “non-utility-based” justification of the welfare criterion in Eq. (1) has a counterpart in multi-commodity settings. Income is replaced by a measure of access to opportunity sets. Concavity of the numerical representation of preferences is ensured by the least joint concave transformation of the opportunity-equivalent well-being functions. Social inequality aversion may be added. More importantly, Theorem 1

¹⁸The set Φ^C is uniquely identified by the opportunity mapping $C \in \mathcal{C}$ and individuals' preferences; no further ethical choice is involved.

shows that the existence of cardinal and interpersonally comparable representations of individuals' preferences need not be exogenously assumed, but can be derived from fairness principles.

Before illustrating the proof, it is worthwhile presenting the formulation of the criteria that emerges when preferences and opportunity sets have a particularly simple form. A preference relation R_i is *homothetic* if for each $\alpha > 0$ and each pair $c, c' \in X$, $c I c'$ implies $\alpha c I \alpha c'$. Let $\mathcal{V}_i$ be the set of numerical representations of preferences R_i that are homogeneous of degree 1. The proof of the existence of such representations is immediate and omitted. An opportunity mapping $C \in \mathcal{C}$ is *homothetic* if for each pair $\lambda, \lambda' > 0$, there exists $\alpha > 0$ such that $c \in C(\lambda)$ if and only if $\alpha c \in C(\lambda')$.

THEOREM 2: *Let preferences be homothetic. Assume there exist a homothetic opportunity mapping $C \in \mathcal{C}$, $\phi \in \Phi^C$, and an increasing and concave function f such that the social ranking $\succsim$ can be represented by the function $W : X^n \rightarrow \mathbb{R}$, defined by setting, for each $x \in X^n$,*

$$W(x) = \sum_{i \in N} f \circ \phi \circ U_i^C(x_i).$$

For each $i \in N$ and some $\lambda^ > 0$, let $z_i \in m_i(C(\lambda^*))$ and $V_i \in \mathcal{V}_i$. Then, $\succsim$ can be represented by the function $\bar{W} : X^n \rightarrow \mathbb{R}$, defined by setting, for each $x \in X$,*

$$\bar{W}(x) = \sum_{i \in N} f \circ \frac{V_i(x_i)}{V_i(z_i)}.$$

The proof is in the [Appendix](#). Intuitively, when preferences are homothetic, any least concave representation is homogeneous of degree 1. Moreover, by homotheticity of the opportunity mapping, an individual's preferred bundle from one opportunity set is proportional to (one of) her preferred bundle from any other opportunity set. As a consequence, interpersonal comparability and cardinality can be simply recovered by normalizing each individual i 's function V_i with respect to the well-being achieved at a reference opportunity set, here $C(\lambda^*)$. Thus, for each individual $i \in N$, this normalized well-being substitutes $\phi \circ U_i^C(x_i)$ in the representation of the *opportunity-equivalent utilitarian* social ranking (up to a positive affine transformation).

3.3. Properties of the Social Ranking

Here, we show that *opportunity-equivalent utilitarian* social rankings satisfy the introduced axioms, thus proving the “if” part of Theorem 1.

LEMMA 2: *If a social ranking is opportunity-equivalent utilitarian, then it satisfies efficiency, continuity, separability, possibility of trade-offs, nondiscrimination, and equal-preference transfer.*

PROOF: Let $\succsim$ be an *opportunity-equivalent utilitarian* social ranking. Let $C \in \mathcal{C}$ and let W represent $\succsim$. Verifying the first three axioms is immediate. By construction, W is increasing with respect to some numerical representation of each individual's preferences. Thus, it satisfies *efficiency*. Continuity of the functions f , ϕ , and $(U_i^C)_{i \in N}$ ensures *continuity*. Furthermore, W is additive with respect to some numerical representation of each individual's preferences. Thus, *separability* holds.

We next show that *possibility of trade-offs* is satisfied. Let $i \in N$ and $x_i \in X \setminus \{0\}$. By construction of U_i^C , individual i is indifferent between x_i and her preferred bundle in $C(U_i^C(x_i))$, that is, $x_i I_i m_i(C(U_i^C(x_i)))$. Let $j \in N \setminus \{i\}$ and $x_j \in m_j(C(U_i^C(x_i)))$. Then, $x_j \in X \setminus \{0\}$ and $U_j^C(x_j) = U_i^C(x_i)$. Moreover, for each $k \in N$, $U_k^C(0) = 0$. Thus, $W(e^i(x_i)) = f \circ \phi \circ U_i^C(x_i) = f \circ \phi \circ U_j^C(x_j) = W(e^j(x_j))$, equivalently, $e^j(x_j) \sim e^i(x_i)$.

To show that $\succsim$ satisfies *nondiscrimination*, let $i \in N$ and $x_i^* \in X$. By construction, $U_i^C(x_i^*)$ is such that $z_i \in m_i(C(U_i^C(x_i^*)))$ satisfies $z_i I_i x_i^*$. Let $X_i \subset X$ be the set of bundles $x_i \in X$ such that $x_i I_i x_i^*$. *Nondiscrimination* is violated if for each $x_i \in X_i$, there exist $j \neq i$ and $x_j < x_i$ such that $e^j(x_j) \succsim e^i(x_i)$. This is impossible as $e^j(x_j) \succsim e^i(z_i)$ with $x_j < z_i$ requires x_j to be j 's preferred alternative in an opportunity set that contains z_i , contradicting strong monotonicity of preferences.

Finally, the concavity of $f \circ \phi \circ U_i^C$ for each $i \in N$ implies that $\succsim$ satisfies *equal-preference transfer*. Formally, consider a progressive transfer between individuals j and k such that $R_j = R_k \equiv R_0$. Before the transfer, j and k are assigned $x_j \in X$ and $x_k \in X$ with $x_j P_0 x_k$. Let $\alpha \geq 0$ be such that, after the transfer, j and k are assigned $x'_j = x_j - \alpha(x_j - x_k)$ and $x'_k = x_k + \alpha(x_j - x_k)$ with $x'_j R_0 x'_k$. Let the remaining individuals be assigned the same bundles at x and x' ; by additivity of W , their assignments are irrelevant for the ranking of x and x' . By definition of concavity,

$$\begin{aligned}\phi \circ U_0^C(x'_j) &\geq (1 - \alpha)\phi \circ U_0^C(x_j) + \alpha\phi \circ U_0^C(x_k), \\ \phi \circ U_0^C(x'_k) &\geq \alpha\phi \circ U_0^C(x_j) + (1 - \alpha)\phi \circ U_0^C(x_k).\end{aligned}$$

Taking the sum, it follows that $W(x') \geq W(x)$, equivalently, $x' \succsim x$.

Q.E.D.

3.4. From the Axioms to the Social Ranking

In this subsection, we illustrate the welfare implications of the axioms. In the first step, we show the (well-known) implication of *efficiency*, *continuity*, and *separability*. At each successive step, we introduce an additional fairness axiom. When all axioms are introduced (Step 4), the social ranking belongs to the class of *opportunity-equivalent utilitarian* criteria, thus proving the “only if” part of Theorem 1.

STEP 1: *If a social ranking $\succsim$ satisfies efficiency, continuity, and separability, then there exists a continuous function $W : X^n \rightarrow \mathbb{R}$ and for each $i \in N$, a representation $\bar{U}_i : X \rightarrow \mathbb{R}$ of her preferences R_i , such that for each pair $x, x' \in X^n$, $x \succsim x'$ if and only if*

$$W(x) \equiv \sum_{i \in N} \bar{U}_i(x_i) \geq \sum_{i \in N} \bar{U}_i(x'_i) \equiv W(x').$$

PROOF: By *continuity*, there exists a continuous function $\bar{W} : X^n \rightarrow \mathbb{R}$ that represents $\succsim$. By *efficiency*, there exists a continuous function $\bar{W} : \mathbb{R}^n \rightarrow \mathbb{R}$ increasing in each argument and for each $i \in N$, a continuous function $V_i : X \rightarrow \mathbb{R}$ representing preferences R_i such that for each $x \in X^n$, $\bar{W}(x) = \bar{W}(V_1(x_1), \dots, V_n(x_n))$. By *separability* (and the assumption that $n \geq 3$), there exists an increasing function $H : \mathbb{R} \rightarrow \mathbb{R}$ and for each $i \in N$, continuous and increasing functions $h_i : \mathbb{R} \rightarrow \mathbb{R}$ such that $\bar{W}(x) = H(\sum_{i \in N} (h_i \circ V_i)(x_i))$. For each $i \in N$, define $\bar{U}_i \equiv h_i \circ V_i$; by construction, this is a numerical representation of preferences R_i . Let $W : X^n \rightarrow \mathbb{R}$ be such that for each $x \in X^n$, $W(x) \equiv \sum_{i \in N} U_i(x_i)$. Since H is increasing, W also represents the social ranking $\succsim$, proving the result. *Q.E.D.*

STEP 2: If the social ranking $\succsim$ also satisfies possibility of trade-offs, then for each pair $i, j \in N$, the numerical representations of individual preferences have the same image (up to a change of origin), that is, there exists $\alpha \in \mathbb{R}$ such that $\bar{U}_i(X) = \bar{U}_j(X) + \alpha$.

PROOF: Assume this is not the case. By continuity, the images of $\bar{U}_i$ and $\bar{U}_j$ are connected subsets of the reals. Define $\underline{u}_i \equiv \inf_{x_i \in X} \bar{U}_i(x_i)$, $\bar{u}_i \equiv \sup_{x_i \in X} \bar{U}_i(x_i)$, $\underline{u}_j \equiv \inf_{x_j \in X} \bar{U}_j(x_j)$, and $\bar{u}_j \equiv \sup_{x_j \in X} \bar{U}_j(x_j)$. Thus, the claim is violated if either $\bar{u}_i - \underline{u}_i > \bar{u}_j - \underline{u}_j$ or the converse. Without loss of generality and rearranging, assume $\bar{u}_i + \underline{u}_j > \bar{u}_j + \underline{u}_i$. By continuity of $\bar{U}_i$, there exists $x'_i \in X$ such that $\bar{u}_i + \underline{u}_j > \bar{U}_i(x'_i) + \underline{u}_j > \bar{u}_j + \underline{u}_i$. Since $\bar{u}_j \equiv \sup_{x_j \in X} \bar{U}_j(x_j)$, there exists no $x'_j \in X$ such that $W(e^i(x'_i)) = W(e^j(x'_j))$, violating possibility of trade-offs. Q.E.D.

Notice that this imposes no restrictions on whether these functions are bounded (above, below, both) or not. It shows that there exists a change of origin such that the images of the numerical representations of individual preferences are the same. For each $i \in N$, let $U_i \equiv \bar{U}_i(X) + \alpha_i$ with $\alpha_i \in \mathbb{R}$ be such that, for each pair $i, j \in N$, $U_i(X) = U_j(X)$.

STEP 3: If the social ranking $\succsim$ also satisfies nondiscrimination, then there exists an opportunity mapping $C \in \mathcal{C}$ and a continuous and increasing function $g: \mathbb{R} \rightarrow \mathbb{R}$ such that for each $i \in N$, $U_i = g \circ U_i^C$. The social ranking $\succsim$ can be represented by $W = \sum_{i \in N} g \circ U_i^C$.

PROOF: For each $i \in N$ and each $x_i \in X$, let the lower- and upper-contour-sets of i at x_i be defined as $\text{LCS}_i(x_i) \equiv \{c \in X \mid x_i R_i c\}$ and $\text{UCS}_i(x_i) \equiv \{c \in X \mid c R_i x_i\}$.

Let $x_i \in X \setminus \{0\}$. For each $\lambda \geq 0$, let $x_i(\lambda) \in X$ be such that $x_i(\lambda) = \lambda x_i$. By possibility of trade-offs, for each $j \in N \setminus \{i\}$ and each $\lambda \geq 0$, there exists $x_j \in X \setminus \{0\}$ such that $e^i(x_i(\lambda)) \sim e^j(x_j)$. Let then

$$\text{LCS}(\lambda) \equiv \left\{ c \in X \mid c \in \bigcap_{j \in N} \text{LCS}_j(x_j) \right\}.$$

We show next that if for some $j \in N$ and some $\lambda \geq 0$, $\text{UCS}_j(x_j(\lambda)) \cap \text{LCS}(\lambda) = \emptyset$, then nondiscrimination is violated. If this is the case, for each $\bar{x}_j \in X$ such that $\bar{x}_j I_j x_j(\lambda)$, there exist $k \in N \setminus \{j\}$ and $\bar{x}_k \in X$ with $\bar{x}_k < \bar{x}_j$ and $\bar{x}_k I_k x_k(\lambda)$. By Pareto indifference (implied by efficiency and continuity), $e^j(\bar{x}_j) \sim e^j(x_j(\lambda))$ and $e^k(\bar{x}_k) \sim e^k(x_k(\lambda))$. By construction, $e^j(x_j(\lambda)) \sim e^k(x_k(\lambda))$. By transitivity, $e^j(\bar{x}_j) \sim e^k(\bar{x}_k)$. This is a violation of nondiscrimination. Thus, for each $j \in N$ and each $\lambda \geq 0$, $\text{UCS}_j(x_j(\lambda)) \cap \text{LCS}(\lambda) \neq \emptyset$. And for each $j \in N$ and each $\lambda \geq 0$, there exists $\hat{x}_j(\lambda) \in X$ such that $\hat{x}_j(\lambda) \in m_j(\text{LCS}(\lambda))$. Then, $x_j(\lambda) I_j \hat{x}_j(\lambda)$ and, by Pareto indifference, $e^j(x_j(\lambda)) \sim e^j(\hat{x}_j(\lambda))$.

For each $\lambda \geq 0$, let $c(\lambda) \in X$ be the smallest bundle such that for each $j \in N$, there exists $\hat{x}_j(\lambda) \in m_j(\text{LCS}(\lambda))$ with $\hat{x}_j(\lambda) \leq c(\lambda)$. Define $C(\lambda) \equiv \{c \in \text{LCS}(\lambda) \mid c \leq c(\lambda)\}$. By construction, $C(\lambda)$ is nonempty, compact, and satisfies free disposal. By continuity of preferences, C is continuous with respect to λ . Moreover, $C(0) = \{0\}$ and $\lim_{\lambda \rightarrow \infty} C(\lambda) = X$. By monotonicity of preferences, $\lambda > \lambda'$ implies that for each $c' \in C(\lambda')$, there exists $c \in C(\lambda)$ with $c \gg c'$, that is, C is strongly monotonic. Thus, $C \in \mathcal{C}$.

For each pair $i, j \in N$ and each pair $x_i, x_j \in X$, $e^i(x_i) \succsim e^j(x_j)$ if and only if $U_i^C(x_i) \geq U_j^C(x_j)$. Together with the results of Steps 1 and 2, this implies that interpersonal comparisons of well-being take place in terms of the opportunity-equivalent well-being function relative to C . Thus, there exists a continuous and increasing function $g: \mathbb{R} \rightarrow \mathbb{R}$

such that for each $i \in N$, $U_i = g \circ U_i^C$. This implies that $\succsim$ can be represented by $W = \sum_{i \in N} g \circ U_i^C$. *Q.E.D.*

STEP 4: If the social ranking $\succsim$ also satisfies equal-preference transfer, then there exists an increasing and concave function $f: \mathbb{R} \rightarrow \mathbb{R}$ and a least joint concave transformation $\phi \in \Phi^C$ such that $g = f \circ \phi$. The social ranking $\succsim$ can be represented by $W = \sum_{i \in N} f \circ \phi \circ U_i^C$ and is opportunity-equivalent utilitarian.

PROOF: Let $j, k \in N$ be such that $R_j = R_k \equiv R_0$. By Step 3, the opportunity-equivalent representation of preferences of j and k are identical, that is, $U_j^C = U_k^C \equiv U_0^C$. Let $x, x' \in X^n$ be such that: (i) $x'_j = x'_k = \frac{x_j + x_k}{2}$; (ii) for each $i \in N \setminus \{j, k\}$, $x_i = x'_i$. By equal-preference transfer, $x' \succsim x$. By the previous steps, this is equivalent to

$$g \circ U_0^C \left(\frac{x_j + x_k}{2} \right) \geq \frac{g \circ U_0^C(x_j) + g \circ U_0^C(x_k)}{2}.$$

Since this holds for each pair of bundles assigned to i and j , $g \circ U_0^C$ is concave. As the argument holds for each pair of individuals with the same preferences, for each $i \in N$, $g \circ U_i^C$ is concave. Thus, $g \in \Psi^C$. By Lemma 1, there exists an increasing and concave function $f: \mathbb{R} \rightarrow \mathbb{R}$ and a least joint concave transformation $\phi \in \Phi^C$ such that $g = f \circ \phi$. *Q.E.D.*

4. CONCLUDING REMARKS

This paper characterizes a family of welfare criteria that: (i) compare alternatives by the sum of indices of each individual's well-being; (ii) respect individuals' preferences; and (iii) rely uniquely on ordinal information of individuals' preferences. Our main contribution is to single out such well-being indices based on fairness arguments.

Two fairness principles define how society makes interpersonal comparisons of well-being: *possibility of trade-offs* limits the influence of each individual in pairwise comparisons; and *nondiscrimination* says that no individual is considered more deserving than any other. These principles force society to choose a family of nested opportunity sets, that is, an opportunity mapping. This choice establishes interpersonal comparability: when individuals are assigned their most preferred bundles from the same opportunity set, they are considered equally well off; when one is assigned her most preferred bundle from a larger opportunity set, she is considered better off. This way of constructing interpersonal comparability includes as special cases the "money metric" and "ray" representations of preferences (Samuelson and Swamy (1974), Pazner (1979)).

The choice of the opportunity mapping essentially depends on the nature and type of commodities (goods, capabilities, rights, etc.) and cannot be determined in our abstract setting. Following the literature on fair allocation theory, one could adopt an axiomatic approach: the axioms would single out a choice rule and associate an opportunity mapping to each specific resource distribution problem (in some domain). An alternative is to argue for specific moral views about the importance of different commodities for individuals' quality of life, for example, building on Rawls (1971)'s primary goods or Sen (1979)'s capabilities.

A further fairness principle ensures that individuals' indices of well-being are cardinal. *Equal-preference transfer* is a generalization of Dalton's transfer principle to multi-commodity spaces. It introduces inequality aversion among individuals with the same preferences and forces the representations of preferences to be concave.

Together with *efficiency*, *continuity*, and *separability*, these fairness principles characterize the *opportunity-equivalent utilitarian* criteria and prove that some versions of “utilitarianism” need not be divorced from the ordinalist foundations of microeconomics.

APPENDIX A: PROOFS

Proof of Lemma 1

PROOF: Let $C \in \mathcal{C}$. By assumption, for each $i \in N$ there exists a concave function $u_i : X \rightarrow \mathbb{R}$ that represents individual i 's preferences R_i . Since U_i^C represents preferences R_i , there exists a real-valued function ψ_i such that $u_i = \psi_i \circ U_i^C$. By concavity of each u_i and finiteness of N , there exists a common real-valued function ψ that, for each $i \in N$, is at least as concave as ψ_i . Thus, $\psi \in \Psi^C$ and Ψ^C is nonempty. Then, the existence of least joint concave functions, that is, $\Phi^C \neq \emptyset$, follows directly from Debreu (1976). *Q.E.D.*

Proof of Theorem 2

PROOF: Let $i \in N$, $x_i \in X$, and $\alpha > 0$. By definition of homogeneity of degree 1, $V_i(\alpha x_i) = \alpha V_i(x_i)$. As $V_i(\alpha x_i)$ is linear with respect to α , V_i is a least concave representation of R_i . Let ϕ_i be a real-valued and strictly increasing function such that for each $x_i \in X$, $\phi_i \circ U_i^C(x_i) = \frac{V_i(x_i)}{V_i(z_i)}$. Since preferences and the opportunity mapping are homothetic, for each $\alpha \geq 0$ there exists $\lambda \geq 0$ such that $\alpha z_i \in m_i(C(\lambda))$. Thus, $\phi_i \circ U_i^C(\alpha z_i) = \frac{V_i(\alpha z_i)}{V_i(z_i)}$ implies that, for each $\alpha \geq 0$, $\phi_i \circ \lambda = \alpha$. As this holds for each $i \in N$, ϕ_i is equal across individuals: let $\bar{\phi} \equiv \phi_i$. Since for each $i \in N$, $\bar{\phi} \circ U_i^C(x_i) = \frac{V_i(x_i)}{V_i(z_i)}$ and $\frac{V_i(x_i)}{V_i(z_i)}$ is a least concave representation of preferences, $\bar{\phi} \in \Phi^C$. As $\phi, \bar{\phi} \in \Phi^C$, $\bar{\phi}$ is a positive affine transformation of ϕ . Thus, $\bar{W}$ is ordinally equivalent to W and represents $\succsim$. *Q.E.D.*

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